SaaS Revenue & Retention Analysis

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Role: Data Analytics Portfolio Project

Tools: SQL (PostgreSQL / BigQuery), Excel Visualization

This report analyzes a SaaS-style dataset to understand revenue growth, customer concentration, acquisition performance, and subscription value distribution.

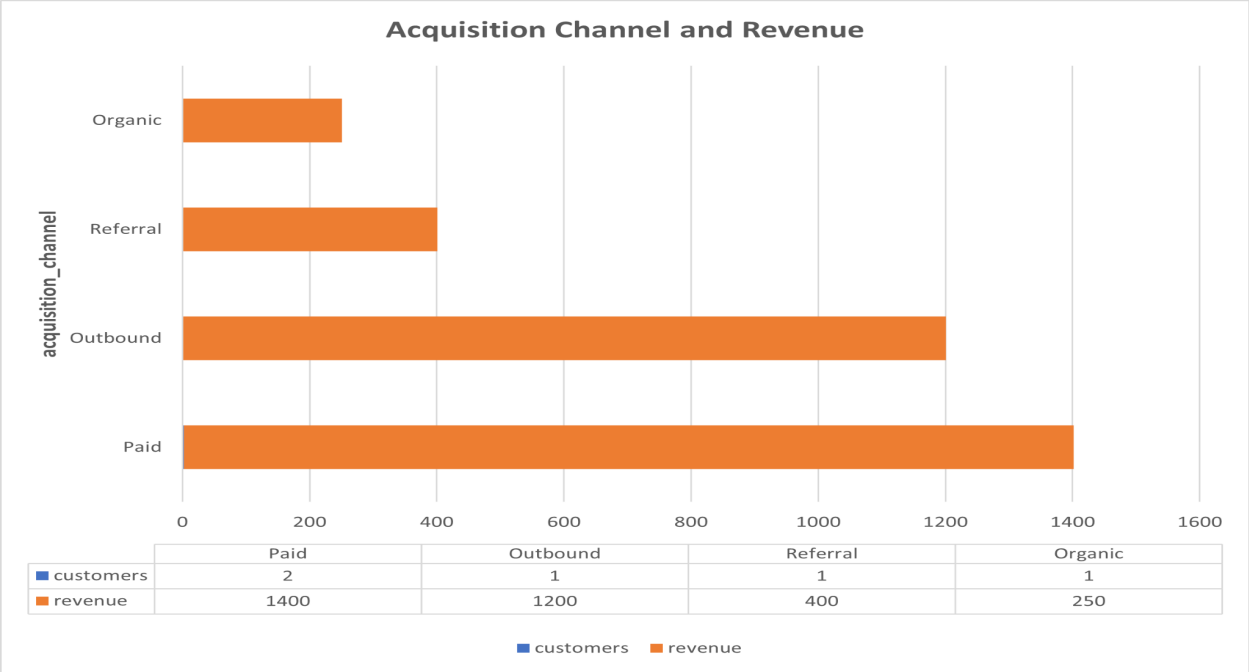
Methodology

The analysis follows a structured analytics workflow similar to consulting-style data analysis: 1. Data Preparation – Clean payment data and filter successful transactions. 2. Revenue Aggregation – Calculate total revenue by marketing channel, month, and customer. 3. Customer Contribution Analysis – Measure revenue share per customer using window functions. 4. Subscription Analysis – Evaluate Monthly Recurring Revenue (MRR) distribution across plans. 5. Cohort Logic – Identify first payment month and track activity over time. SQL queries were used to compute revenue metrics and extract analytical insights.

SQL Query Explanation

Several SQL techniques were used to perform the analysis: • JOIN – Combine customer, payment, and subscription datasets. • GROUP BY – Aggregate revenue by channel, month, and customer. • CTE (Common Table Expressions) – Structure complex queries into readable steps. • Window Functions – Calculate revenue share and cohort metrics. • Date functions – Create monthly revenue trends and cohort indexing. These techniques are commonly used in real SaaS analytics workflows.

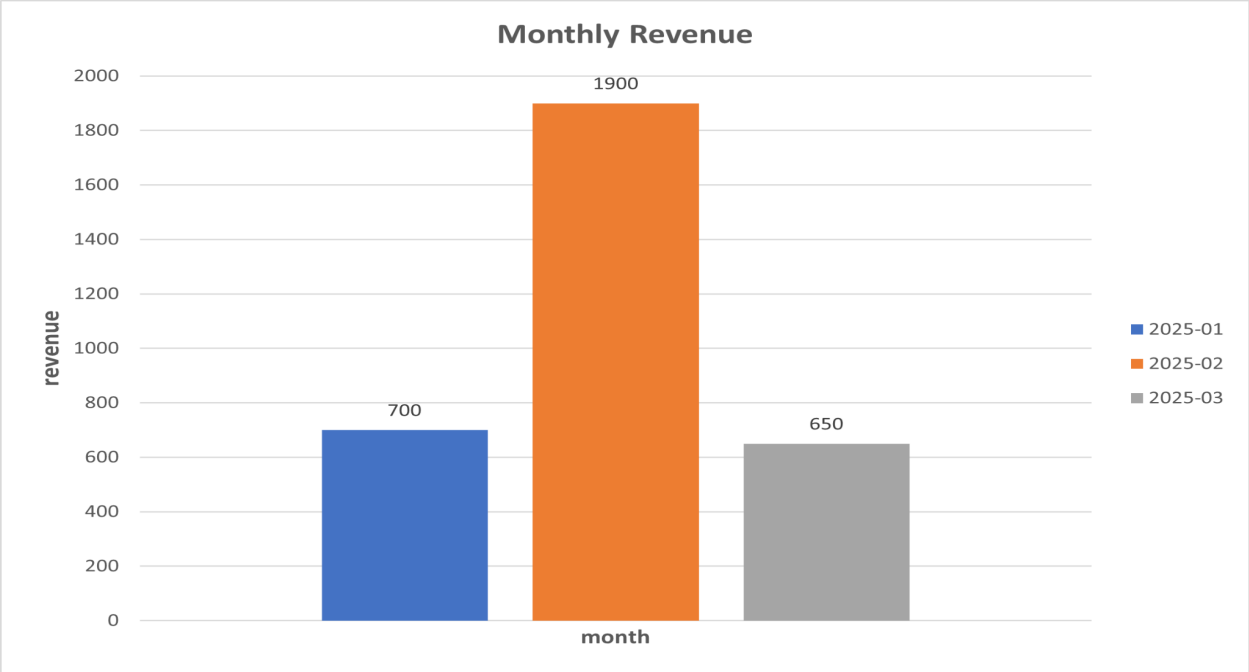
Revenue by Acquisition Channel



	Paid	Outbound	Referral	Organic
customers	2	1	1	1
revenue	1400	1200	400	250

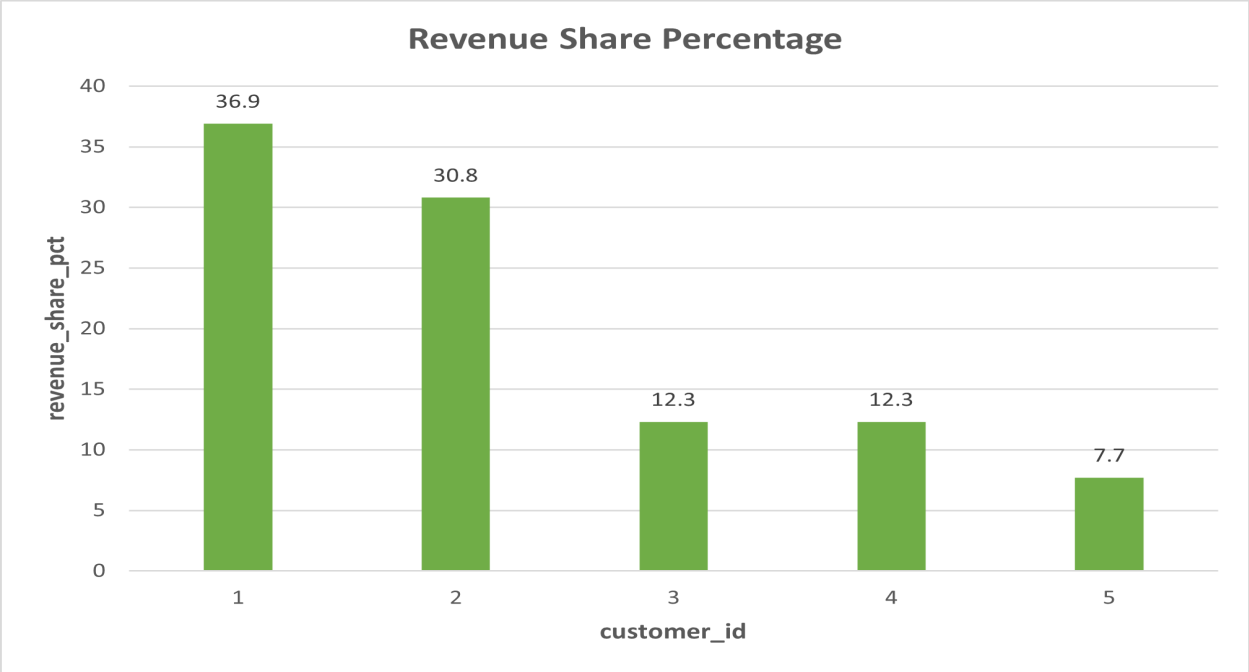
Insight: Paid and Outbound channels generate the majority of revenue, indicating that growth is primarily driven by marketing spend and outbound sales efforts.

Monthly Revenue Trend



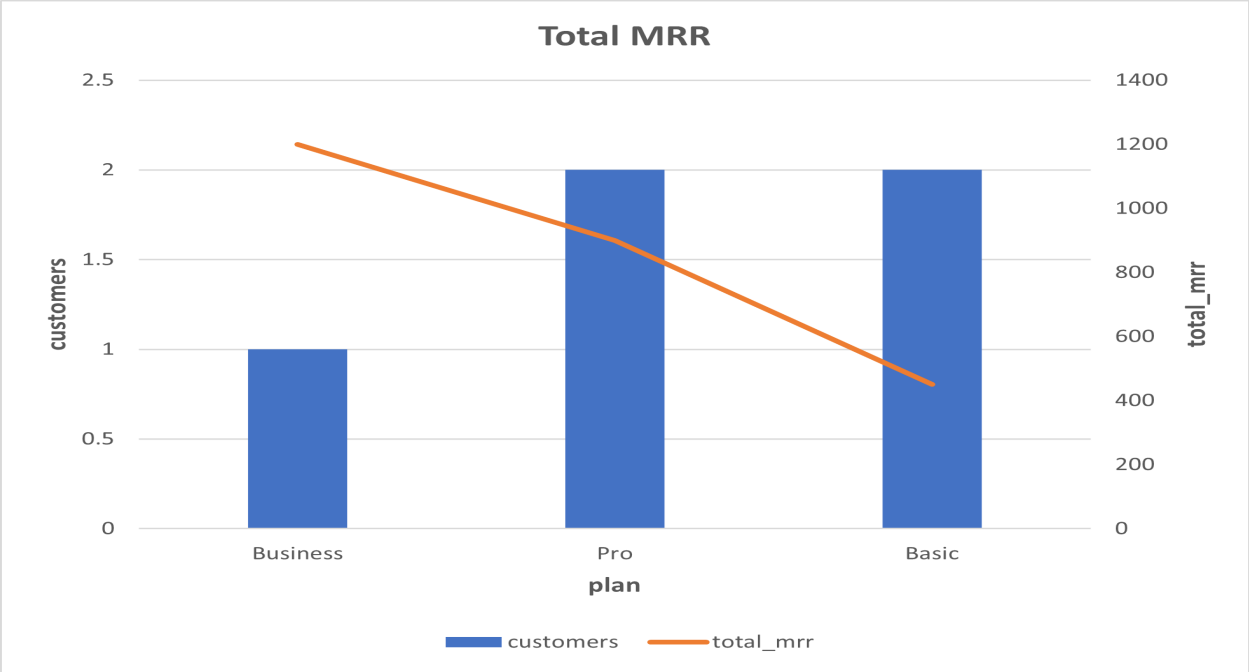
Insight: Revenue peaks in February, suggesting a potential seasonal pattern or the impact of a successful campaign.

Customer Revenue Share



Insight: Revenue is highly concentrated among a small number of customers, highlighting potential revenue concentration risk.

MRR Distribution by Plan



Insight: Higher-tier plans generate significantly more revenue per user, suggesting opportunities for upselling lower-tier customers.

Conclusion

This analysis demonstrates how SQL can be used to answer important business questions in a SaaS environment. Key findings include: • Marketing-driven growth dominated by paid and outbound channels • Revenue volatility across months • Revenue concentration among a small number of customers • Strong revenue contribution from higher subscription tiers These insights could inform future strategy such as improving customer diversification and increasing upsell efforts.